



Find the Best Bonds¹

Search our vast universe of bonds with the free Bond Search Tool to find the best bond prices with no mark-ups

Rated Best
Online Broker 2023
for Bonds
by Benzinga¹



Interactive Brokers
Rated #1 ... Again
Best Online Broker
2023 by Barron's²

The best-informed
investors choose Interactive Brokers



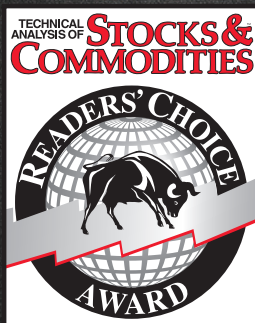
ibkr.com/find-bonds



Member - NYSE, FINRA, SIPC – Supporting documentation for any claims and statistical information will be provided upon request. [1] Rated best online broker for bond brokers according to Benzinga - Best Online Brokers for Bonds, February 22, 2023. [2] Interactive Brokers rated #1, Best Online Broker according to Barron's Best Online Brokers Survey of 2023: June 9, 2023. For more information see, ibkr.com/info - Barron's is a registered trademark of Dow Jones & Co. Inc.

07-IB23-1621CH1599

Can you imagine winning the
Stocks & Commodities Readers' Choice
Award for 30 consecutive years?



*Best Standalone
Analytical Software
in it's price category
1993-2023*



We can.

Get your free trial at MetaStock.com/TASC

This is neither a solicitation to buy or sell any type of financial instruments, nor intended as investment recommendations. All investment trading involves multiple substantial risks of monetary loss. Don't trade with money you can't afford to lose. Trading is not suitable for everyone. Past performance, whether indicated by actual or hypothetical results or testimonials are no guarantee of future performance or success. NO REPRESENTATION IS BEING MADE THAT ANY ACCOUNT WILL OR IS LIKELY TO ACHIEVE PROFITS OR LOSSES SIMILAR TO THOSE SHOWN. IN FACT, THERE ARE FREQUENTLY SHARP DIFFERENCES BETWEEN HYPOTHETICAL PERFORMANCE RESULTS OR TESTIMONIALS AND THE ACTUAL RESULTS SUBSEQUENTLY ACHIEVED BY ANY PARTICULAR TRADING PROGRAM. Furthermore, all internal and external computer and software systems are not fail-safe. Have contingency plans in place for such occasions. MetaStock assumes no responsibility for errors, inaccuracies, or omissions in these materials, nor shall it be liable for any special, indirect, incidental, or consequential damages, including without limitation losses, lost revenue, or lost profits, that may result from reliance upon the information presented.

SYSTEM DESIGN

Simplifying Your Life

Moving Average Computations, Made Easier

Presenting programming tricks to simplify coding indicators and strategies.

by John F. Ehlers

A

simple moving average of length N is computed by adding N values and dividing the sum by N . The process is repeated on a bar-by-bar basis. What could be easier? While conceptually simple, coding for long moving averages can quickly become wearisome because there are so many terms.

This tedium can be reduced by putting the summation in a loop, but looping is difficult to achieve in some applications, such as Excel. Another simplifying approach is to drop off the oldest value and add a new value to the moving average. However, this requires computing the initial value of the long moving average at least once. Here are two ways to compute the simple moving average (SMA) with ease.

In Z-transform notation, a unit delay is represented by Z^{-1} . The transfer response is the output of the filter divided by its input. Thus, the transfer response of an eight-bar simple moving average would be written as:

(Equation 1)

$$H(z) = (1 + Z^{-1} + Z^{-2} + Z^{-3} + Z^{-4} + Z^{-5} + Z^{-6} + Z^{-7})/8$$

This same expression, written in EasyLanguage where a delay of N bars is represented in square brackets as $[N]$, is written as equation 2:

(Equation 2)

$$SMA = (Price + Price[1] + Price[2] + Price[3] + Price[3] + Price[4] + Price[5] + Price[6] + Price[7])/8$$

Equation 1 is a simple finite series that can be written most generally in fractional form as where $Y(z)$ is the filter output and $X(z)$ is the filter input:

(Equation 3)

$$H(z) = \frac{Y(z)}{X(z)} = \left(\frac{1 - z^{-N}}{1 - z^{-1}} \right) / (N + 1)$$

Equation 3 is identically equivalent to equation 1 if $N=7$, and is therefore a simple moving average. When we carry out the crossmultiplication of equation 3, we obtain:

(Equation 4)

$$Y(z) = (X(z)(1 - z^{-N}) + Y(z)z^{-1})/(N+1)$$

Equation 4 provides the means to program an arbitrarily long SMA using just a few terms. The EasyLanguage equivalent of equation 4 is:

(Equation 5)

$$SMA = (Price - Price[N] + SMA[1]) / (N+1)$$

Another SMA programming trick can be accomplished by recognizing that we don't have to do the filtering all at once. Rather, we can *cascade* filters. That means we can filter the output of a previous filter that takes the output of a previous filter. Cascading filters is represented by multiplication in Z transforms. The SMA transfer response of cascaded filters can be written as:

(Equation 6)

$$H(z) = \frac{(1 + z^{-1})(1 + z^{-2})(1 + z^{-4}) \dots (1 + z^{-2^{K-1}})}{2^K}$$

For example, if $K=3$, we would have an eight-bar SMA. As a test, we can expand equation 6 to:

(Equation 7)

$$\begin{aligned} H(z) &= (1 + z^{-1})(1 + z^{-2})(1 + z^{-4}) / 8 \\ &= (1 + z^{-1} + z^{-2} + z^{-3})(1 + z^{-4}) / 8 \\ &= (1 + z^{-1} + z^{-2} + z^{-3} + z^{-4} + z^{-5} + z^{-6} + z^{-7}) / 8 \end{aligned}$$

Thus, equation 7 shows that the cascaded filters expand to be identical with an SMA. In EasyLanguage, the cascaded filters would be written as:

$$\begin{aligned} Value1 &= Price + Price[1] \\ Value2 &= Value1 + Value1[2] \\ Value3 &= Value2 + Value2[4] \\ SMA &= Value3 / 8 \end{aligned}$$

These programming tricks should simplify the coding of your indicators and strategies.

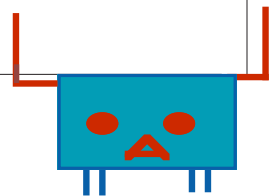
A pioneer in introducing maximum entropy spectral analysis to technical analysis, John Ehlers is president of MESA Software. Watch for his new book, Cybernetic Analysis For Stocks And Commodities, coming in March 2004.

SUGGESTED READING

Ehlers, John F. [2001]. *Rocket Science For Traders*, John Wiley & Sons.

INDICATORS

These programming tricks should simplify the coding of your indicators and strategies.





Traders take many paths to reach their destination.
Can your platform get you there?

NINJA**TRADER** ECOSYSTEM

Search apps and services to personalize the
NinjaTrader platform to meet your requirements.
Indicators, automated strategies, free tools & more.

Explore now at ninjatraderecosystem.com

Futures, foreign currency and options trading contains substantial risk and is not for every investor. Only risk capital should be used for trading and only those with sufficient risk capital should consider trading.

TRADE BRILLIANTLY



thinkorswim® is now at Schwab.

Our award-winning thinkorswim trading platforms are loaded with powerful features that let you dive deeper into the market.

- Visualize your trades in a new light on thinkorswim desktop with robust charting and analysis tools, including 400+ technical studies.
- Uncover new opportunities with up-to-the-minute market news and insights.
- Choose a platform to fit your trading style—from streamlined to advanced.
- Available on desktop, web, and mobile to meet you where you are so you never miss a thing.

Built by the trading-obsessed, so you can trade brilliantly.



Investing involves risks, including loss of principal.
Schwab does not recommend the use of technical analysis as a sole means of investment research.
© 2023 Charles Schwab & Co., Inc. All rights reserved. Member SIPC. (1023-373U) ADP121824-00

[Schwab.com/Trading](https://www.schwab.com/trading)

SUBSCRIBE OR RENEW TODAY!

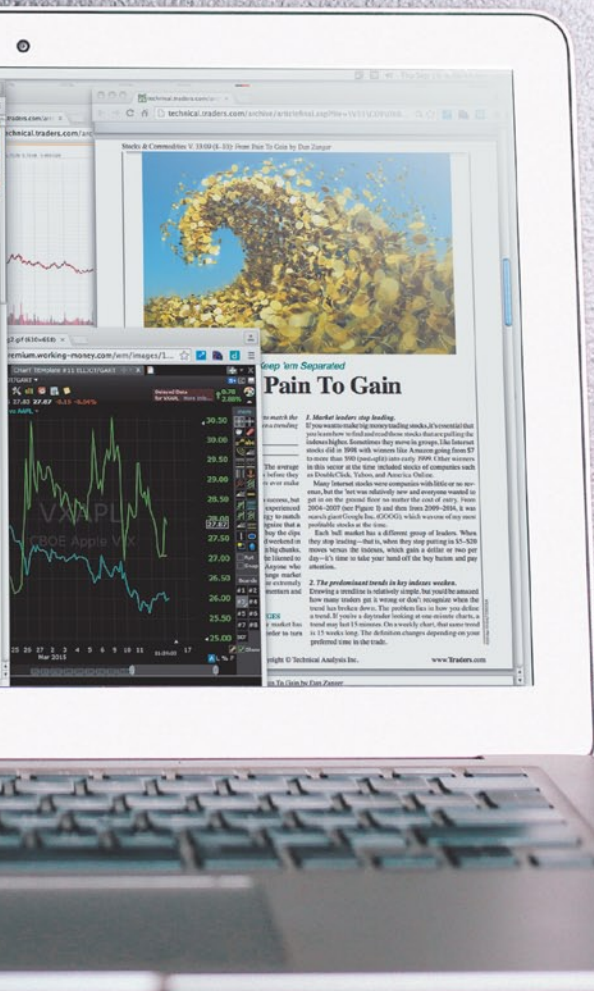
Every Stocks & Commodities subscription (regular and digital) includes:

- Full access to our Digital Edition
The complete magazine as a PDF you can download.
- Full access to our Digital Archives
That's 35 years' worth of content!
- Complete access to WorkingMoney.com
The information you need to invest smartly and successfully.
- Access to Traders.com Advantage
Insights, tips and techniques that can help you trade smarter.

1 year **\$89⁹⁹**

2 years **\$149⁹⁹**

3 years **\$199⁹⁹**



PROFESSIONAL TRADERS' STARTER KIT

A 5-year subscription to S&C magazine that includes everything above PLUS a free* book, *Charting The Stock Market: The Wyckoff Method*, all for a price that saves you \$150 off the year-by-year price! *Shipping & handling charges apply for foreign orders.

5 years **\$299⁹⁹**

That's around \$5 a month!



Visit **www.Traders.com** to find out more!